Problem Set 4 Fin Econ

1.a.

$$R_f = \frac{U'(C_t)}{\beta E_t U'(C_{t+1})} = \frac{C_0^{-\alpha}}{\frac{4}{10}\left(\frac{1}{3}\cdot C_1(w_1)^{-\alpha} + \frac{1}{3}(C_2(w_2)^{-\alpha} + \frac{1}{3}(C_3(w_3)^{-\alpha}\right.}$$

$C_0 = e_0 = 1$

$C_1(w_1) = 3 \quad C_1(w_2) = 2 \quad C_1(w_3) = 1$

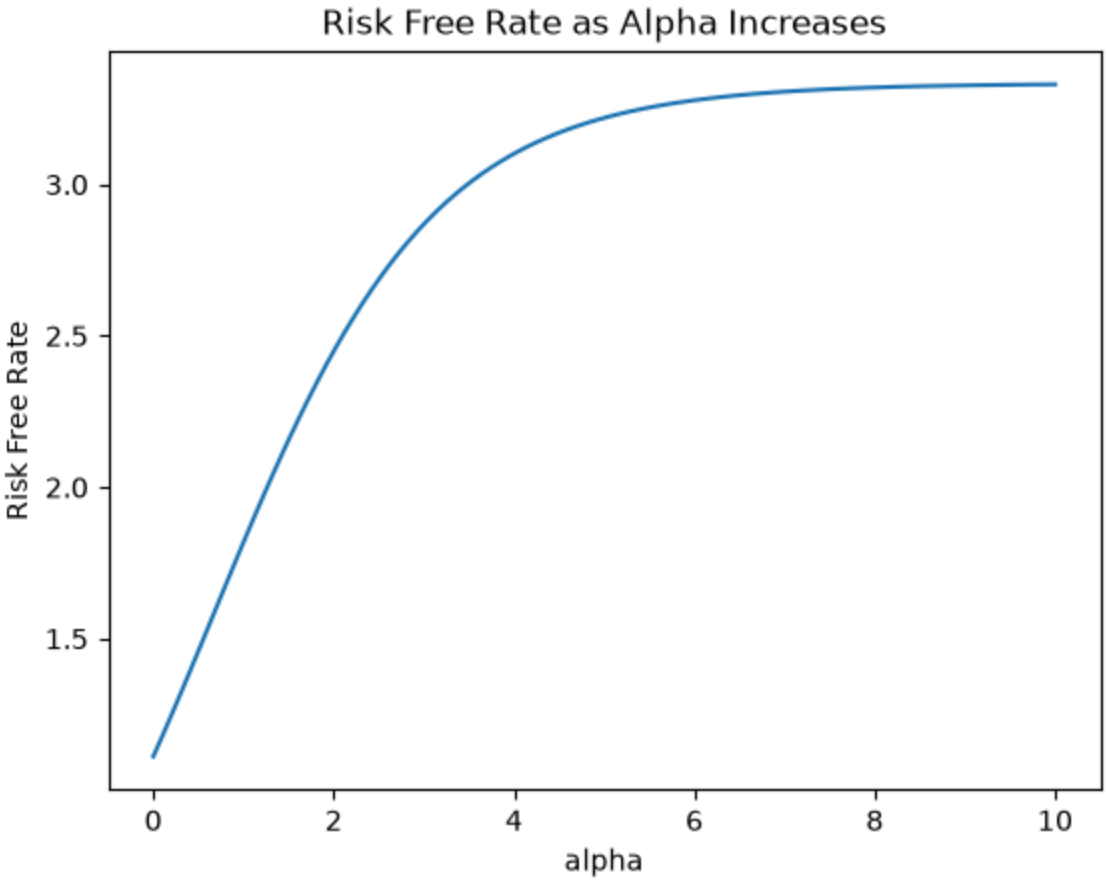
$$R_f = \frac{1^{-\alpha}}{\frac{4}{10}\left(\frac{1}{3}\cdot 3^{-\alpha} + \frac{1}{3}\cdot 2^{-\alpha} + \frac{1}{3}\cdot 1^{-\alpha}\right)} = \frac{10}{3}\cdot\frac{1}{\left(3^{-\alpha}+2^{-\alpha}+1\right)}$$

Problem Set 4 Code

In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

Problem 1(a)

In [3]:
```
alpha = np.linspace(0,10, 100)
rf = 10/3 * 1/ (1+ 3**(-alpha)+2**(-alpha))
plt.plot(alpha,rf)
plt.xlabel('alpha')
plt.ylabel('Risk Free Rate')
plt.title('Risk Free Rate as Alpha Increases')
plt.show()
```



as alpha increases our risk free rate increases but at a decreasing rate. Alpha is our risk aversion, thus we would prefer to smooth our consumption over time. Our elasticity of intertemporal consumption decreases, thus we prefer not to change our consumption as much. We need a higher risk free rate to change our consumption patterns at our given risk aversion.

1.6

market clearing

$$R_t = \frac{D_t + P_t}{P_{t-1}} \qquad P_t = B \cdot E_0\left(\frac{U'(C_1)}{U'(C_0)} \cdot C_1\right) \leftarrow C_t = D_t$$

$$= \frac{9}{10}\left(\frac{1}{3}\left(2^{-\alpha}\cdot 2 + 2^{-\alpha}\cdot 2 + 1\cdot 1^{-\alpha}\right)\right.$$

$$= \frac{3}{10}\left(3\cdot 3^{-\alpha} + 2^{-\alpha} + 1\right)$$

$$R_1 = \frac{D_1}{P_0} \qquad P_0 = \frac{3}{10}\left(3^{1-\alpha} + 2^{1-\alpha} + 1\right)$$

$$R_1(w_s) = \frac{D_1(w_s)}{P_0}$$

$$E(R_1) - R_f = \frac{1}{3}\cdot\left(\frac{3}{P_0} + \frac{2}{P_0} + \frac{1}{P_0}\right) - \frac{10}{3}\left(\frac{1}{3^{-\alpha} + 2^{-\alpha} + 1}\right)$$

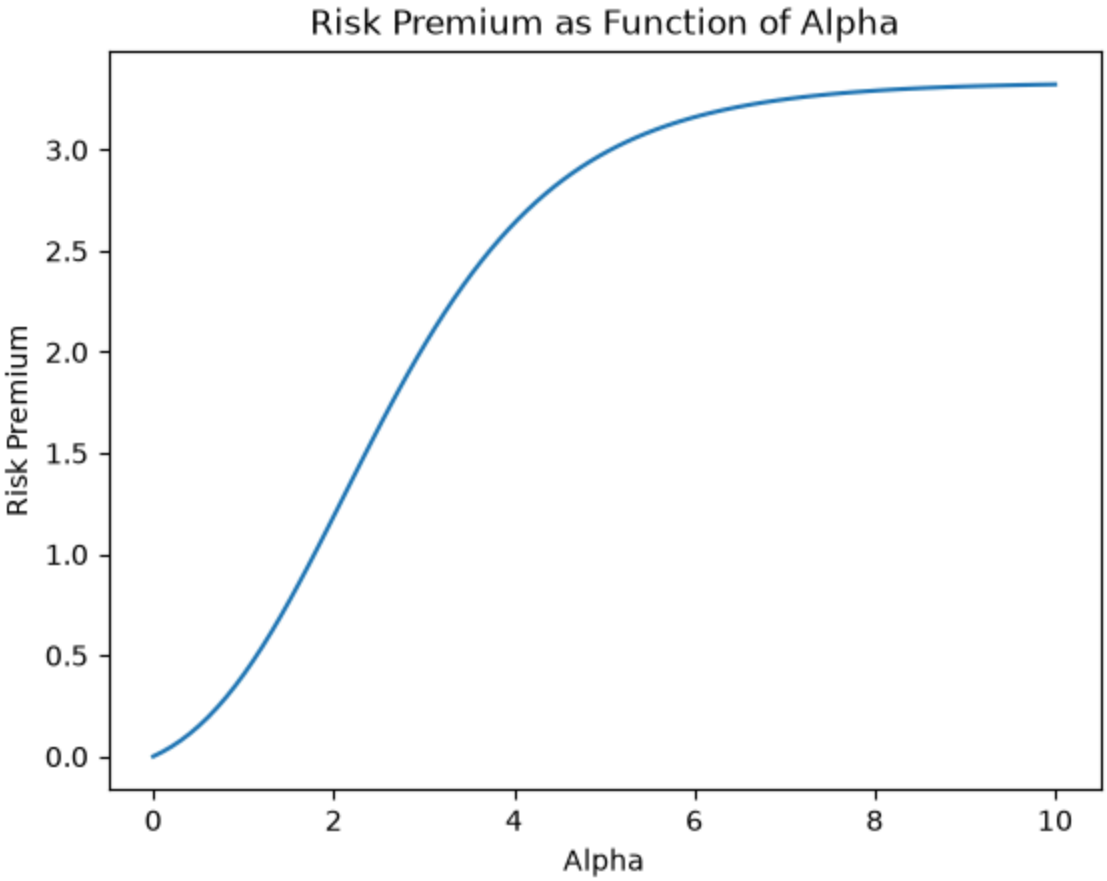
Problem 1(b)

In [4]:
```
p0 = 3/10 * (3**(1-alpha) +2**(1-alpha)+1)

eri = 1/3* (3/p0 +2/p0 +1/p0)

risk_prem = eri-rf
```

In [6]:
```
plt.plot(alpha,risk_prem)
plt.xlabel('Alpha')
plt.ylabel('Risk Premium')
plt.title('Risk Premium as Function of Alpha')
plt.show()
```



As risk aversion increases our risk premium also increases. For the agent to hold the risky asset, the risk premium must increase the more risk averse they are. They need the expected return to be high enough to offset states where consumption will be lower.

2a.

Expected Payoff $= \frac{1}{2}e + \frac{1}{2}(1-a)e$

$p_0 = 0$

$0 = E[U'(C_1) \cdot (D_1 - p)]$ $\quad C_1 = D_1$ market clearing

$$0 = \frac{1}{2}\left(\frac{e-p}{e} + \frac{(1-\alpha)e - p}{(1-\alpha)e}\right) = \frac{(1-\alpha)e - (1-\alpha)p + (1-\alpha)e - p}{(1-a)e}$$

$$0 = \frac{2(1-a)e - p(2-\alpha)}{(1-a)e}$$

$2(1-a)e = p(2-\delta)$ $\quad$ $p = \frac{2(1-a)e}{2-a}$

$$E(l) = \frac{\frac{1}{2}(e + (1-a)e)}{\frac{2(1-a)e}{2-a}} = \frac{\frac{1}{2}(2-a)(e)}{2(1-a)e} \cdot 2-a = \frac{(2-a)^2}{4(1-a)}$$

2b

$e_1(w_1) = e$ $\quad p = 1/2$

$e_1(w_2) = e$ $\quad p = 1/2(1-\delta)$

$e_1(w_3) = (1-a/\delta)e$ $\quad p = 1/2\,\delta$

$C_1 = e_1$ treat as one agent

$0 = E[U'(C_1)(D_1 - p)]$

2b cont

dividend = aggregate endowment

$D_1 = (1-\delta)e + \delta(1-a/\delta)e$

$= e - \delta e + \delta e - ae = (1-a)e$

$e_1 = c_1$ like single agent model

$$0 = \frac{1}{2}\frac{1}{e}\cdot(e-p) + \frac{1-\delta}{2}\cdot\frac{1}{e}\cdot((1-a)e-p) + \frac{\delta}{2}\frac{1}{(1-a/\delta)e}\cdot((1-a)e-p)$$

$$= e-p + (1-\delta)(e-ae-p) + \frac{\delta}{(1-a/\delta)}\cdot(e-ae-p)$$

$$= e-p+e-ae-p-\delta e+\delta ae+\delta p + \frac{\delta^2}{\delta-a}\cdot(e-ae-p)$$

$$= e(1-a-\delta+\delta a) - p(2+\delta) + \frac{\delta^2 e - \delta^2 ae - \delta^2 p}{\delta-a}$$

$$= e(2-a-\delta+\delta a)\cdot(\delta-a) - p(2-\delta)\cdot(\delta-a) + \delta^2 e - \delta^2 ae - \delta^2 p$$

$$= e(2\delta - a - \delta^2 + \delta^2 a - 2a + a^2 + \delta a - \delta a^2 + \delta^2 - \delta^2 a)$$

$$- p(2\delta - 2a - \delta^2 + \delta a + \delta^2)$$

$$= e(2\delta + \delta^2 a - 2a + a^2 - \delta a^2 - \delta^2 a) - p(2\delta - 2a + \delta a)$$

$$p = e\frac{(2\delta - 2a + a^2 - \delta a^2)}{2\delta - 2a + \delta a} \qquad \text{when } \delta=1 \to p = \frac{2(1-a)e}{2-a}$$

Expected payoff $= \frac{1}{2}\cdot e + \frac{1-\delta}{2}e + \frac{\delta}{2}(1-a/\delta)e$

$= \frac{1}{2}e(2-\delta+\delta-a) = \frac{1}{2}e(2-a)$ same as before

$$E(Q) = \frac{\frac{1}{2}e(2-a)\cdot(2\delta-2a+\delta a)}{e(2\delta+2a+a^2-\delta a^2)} = \frac{(2-a)(2\delta-2a+\delta a)}{2(2\delta-2a+a^2-\delta a^2)}$$

2b (cont)

$\delta = 1$ the $E(R) = \frac{(2-a)^2}{4(1-a)}$ same as 2(a).

when $\delta < 1$ then P is greater as $(1-\delta)a^2$ > effect in numerator will be smaller than $(1-\delta)a$ in denominator.

For E(R) when $\delta < 1$ will be lower for same reason as $(1-\delta)a^2$ is in denominator now. $1-\delta = 0$ when $\delta = 1$ which just reverts the problem setting in 2a. If $\delta \to 1$ then we are less likely to get bad endowment if e is high. This means our price will be higher despite and thus lower average returns.